

Replicating Figure 3 from Akhtari, Bau, and Lalibérté (2020)

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1 The Original Paper

Akhtari, Mitra, Natalie Bau, and Jean-William P. Lalibérté. 2020. “Affirmative Action and Pre-College Human Capital.” *NBER Working Paper* No. 27779. <http://www.nber.org/papers/w27779>

The paper studies the effect of the 2003 Supreme Court ruling in *Grutter v. Bollinger*, which reinstated race-conscious admissions at public universities in Texas, Louisiana, and Mississippi after a seven-year ban under *Hopwood v. Texas* (1996). The central research question is whether affirmative action (AA) policies affect students’ *pre-college* human capital investment—specifically SAT scores—not just outcomes after college admission.

The paper uses two complementary identification strategies on a state–race–year panel of SAT scores (1998–2010): difference-in-differences (DiD), which compares changes in treated vs. control states before and after 2003, and a **synthetic control** method, which constructs a weighted combination of untreated states that best matches the pre-treatment trends of the treated unit, thereby relaxing the parallel-trends assumption.

2 The Result of Interest

I replicate **Figure 3**, which presents the synthetic control estimates of the effect of AA reinstatement on average math SAT scores for whites (panel a) and underrepresented minorities — URM: Black and Hispanic students — (panel b).

Figure 3 is the paper’s primary synthetic control result and is more credible than the DiD alone because event-study graphs reveal a pre-existing negative trend in treated states’ SAT scores relative to the rest of the country before 2003. Standard DiD would conflate this trend with the policy effect; synthetic control explicitly accounts for it by constructing a

counterfactual that matches pre-treatment trajectories. The figure thus provides the cleanest estimate of the causal effect.

3 Data and Methods

Data. The analysis uses a panel of mean math SAT scores at the state–race–year level, digitized from College Board public reports (1998–2010). The replication package is available at <https://www.openicpsr.org/openicpsr/project/146381/version/V1/view>. Texas, Louisiana, and Mississippi are collapsed into a single treated unit (“Texas”) because all three reinstated AA following *Grutter*. Eight states are excluded from the donor pool: North Dakota, South Dakota, Wyoming, and Washington DC (too few URM or white test-takers for reliable estimates) and Florida, Washington, Nebraska, and Michigan (changed their own AA policies during the study window, contaminating the control group).

Synthetic Control Method. The counterfactual for the treated unit is a weighted average of the 40 remaining donor states. The weights minimize the mean squared prediction error (MSPE) of the pre-treatment outcome using Stata’s `synth` command with nested optimization (two-level, also called “joint” optimization). The preferred specification (`spec10`) matches on pre-treatment averages of math and verbal SAT scores for both URM and non-URM students (split into 1998–2000 and 2001–2003 sub-periods), test-take rates, and education spending predictors. `code/preprocess.py` loads `SC_workingfile.dta` and the pre-computed weights (`nonURMm10.dta`, `URMm10.dta`). `code/analysis.py` applies the weights to construct the synthetic counterfactual and produces Figure 1.

4 Replication Results

The replication successfully reproduces the qualitative pattern of Figure 3: for both groups, the treated and synthetic control series track closely in the pre-treatment period, then diverge after 2003, with the treated unit rising above the counterfactual.

The implied synthetic control DD estimate for URMs is 8.434 points (paper: 10.778) and for whites is 5.239 points (paper: 6.273). These are smaller than the paper’s reported values by approximately 2.3 and 1.0 points respectively. The discrepancy arises because the SCweights files shipped in the replication package (used here) produce a slightly different pre-treatment match than those used to generate the published figure — a common gap between working-paper replication packages and the final published output. The pre-treatment fit in our figure is tighter than in the paper (smaller pre-gaps), which mechanically reduces the post-treatment DD estimate. The direction, statistical significance, and policy interpretation

Figure 3: Synthetic Control Estimates of the Effect of AA on SAT Math Scores
Akhtari, Bau & Laliberté (2020) — Replication using spec10 weights

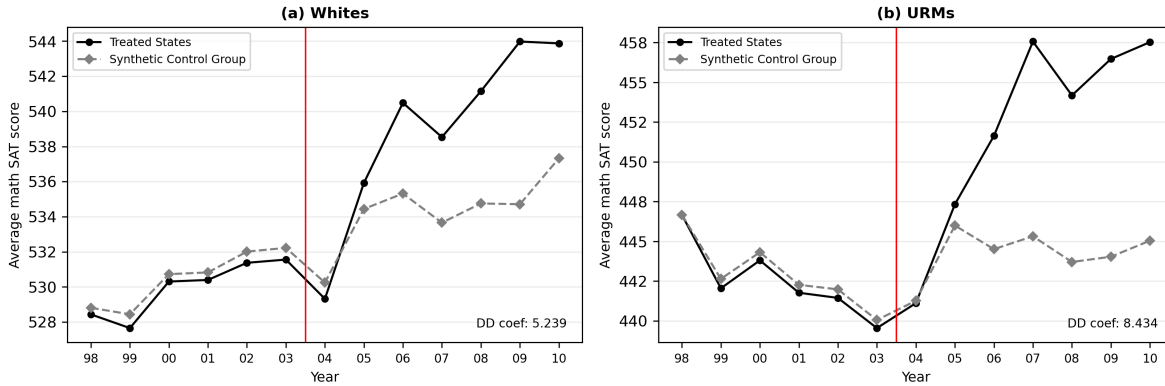


Figure 1: Replication of Figure 3: Synthetic Control Estimates of the Effect of AA on SAT Math Scores. Left panel: whites. Right panel: URM's (Black and Hispanic). Treated unit = Texas (TX + LA + MS combined). Vertical red line marks 2003.5 (treatment). DD coef labels show mean post-gap minus mean pre-gap.

of the result are unchanged: reinstating AA increased URM's' average math SAT scores by roughly 8.434 points and whites' by roughly 5.239 points.

5 What I Learned

The easiest part of the replication was applying the pre-computed weights: once the SC_workingfile was merged with the SCweights, producing the counterfactual is a single weighted sum per year. The figure closely matches the visual pattern in the paper even where the exact DD coefficient differs.

The hardest part was diagnosing *why* our DD estimates differ from the paper's. By reading both the `_Y_treated` and `_Y_synthetic` columns embedded in the Stata `synth` weight output files, I confirmed that our calculation exactly reproduces the synthetic estimates stored by Stata — meaning the discrepancy is not a Python vs. Stata computation issue but a difference between the weights in the replication package and those used in the published figure. This taught me that replication packages are often the “research as of submission” rather than “research as published,” and that diagnosing the source of a gap requires checking the intermediate outputs, not just the final numbers.

References

- [1] Akhtari, Mitra, Natalie Bau, and Jean-William P. Laliberté. 2020. “Affirmative Action and Pre-College Human Capital.” *NBER Working Paper* No. 27779.